CHEMISTRY

SECTION – I
Straight Objective Type

1. **(a)** Precipitation of NaCl will take place after solution become saturated.

Molarity of sample of sea water, $M = \frac{W_{NaCl}}{M_{NaCl}} \times \frac{1000}{V_{in\ mL}} = \frac{2.8}{58.5} \times \frac{1000}{100/1.03} = 0.493\ M$

$M_1V_1 = M_2V_2$

$0.493 \times 4 = 5.45 \times V_2$

$V_2 = 0.361\ L$

Water to be evaporated would be = (4 – 0.361)L = 3.639 L ≃ **3.64 L**.

2. **(c)** The cell is $Pt \mid H_2 \mid H^+(1\ M) \parallel Ag_2SO_4$ (saturated) $\mid Ag(s)$

At anode: $H_2 \longrightarrow 2H^+ + 2e^-$

At cathode: $2Ag^+ + 2e \longrightarrow Ag$

$H_2 + 2Ag^+(aq) \longrightarrow 2H^+(aq) + Ag(s)$

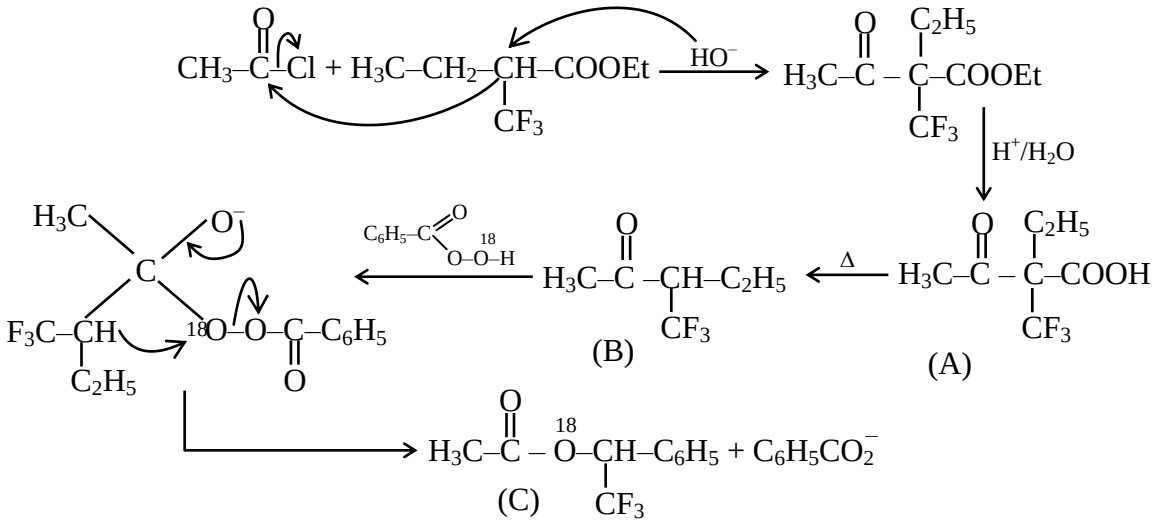
$E_{cell} = E^o_{cell} - \frac{0.059}{2}\log\frac{[H^+]^2}{[Ag^+]^2} \times \frac{1}{p_{H_2}}$

$0.74 - 0.799 = -0.059\ \log\frac{1}{[Ag^+]\times 1}$; $1 = \log\frac{1}{[Ag^+]}$; $[Ag^+] = 10^{-1}\ M$

$Ag_2SO_4(s) \rightleftharpoons 2Ag^+(aq) + SO_4^{2-}(aq)$

$[Ag^+] = 10^{-1}$ and $[SO_4^{2-}] = \left[\frac{10^{-1}}{2}\right]$; $K_{sp} = [Ag^+]^2\ [SO_4^{2-}] = [10^{-1}]^2 \left[\frac{10^{-1}}{2}\right] = \mathbf{5 \times 10^{-4}}$.

3. **(c)**



4. **(c)** Total number of ^{40}K atoms in 80 kg human body

$$= \frac{1.2\times10^{-2}}{100}\times\frac{3.5\times10^{-1}}{100}\times80\times10^{3}\times\frac{6.02\times10^{23}}{40}$$

$= 5.06 \times 10^{20}$ atoms

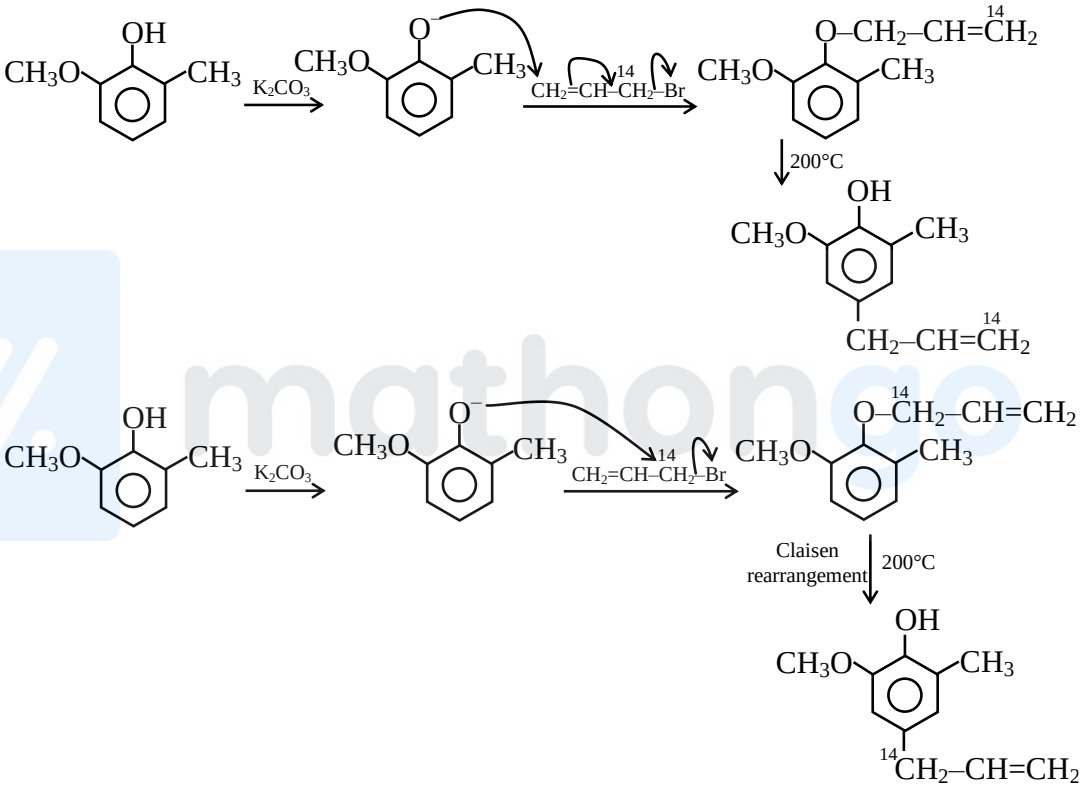
$$\lambda = \frac{0.693}{6.9\times10^{14}} = 1.004 \times 10^{-15} \simeq 10^{-15}.$$

$A = N\lambda$

$= 5.06 \times 10^{20} \times 10^{-15} =$ **5.06×10^{5} dpm**.

SECTION – II
Multiple Correct Choice Type

5. **(a, b)**



6. **(b, c)**

$$Na_2S + Na_2SO_3 + I_2 \longrightarrow \underset{(A)}{Na_2S_2O_3} + 2NaI$$
(B)

$$Na_2S_2O_3 + AgNO_3 \longrightarrow \underset{\text{(white ppt.)}}{Ag_2S_2O_3} + 2NaNO_3$$

$$Ag_2S_2O_3 + H_2O \longrightarrow \underset{\text{(black ppt.)}}{\underset{(C)}{Ag_2S}} + H_2SO_4$$

7. **(b, d)** Vapour pressure of solution is inversely proportional to its boiling point. ∴, boiling point of (A) would be lower than (B), as its vapour pressure is higher than (B).

If (A) and (B) solution have shown positive deviation than it would form minimum boiling point azeotrope.

8. **(a, b, c)**

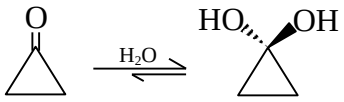
$NH_4Cl + NaNO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

$3CaOCl_2 + 2NH_3 \xrightarrow{\Delta} N_2 + 3CaCl_2 + 3H_2O$

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

$CaCN_2 + 4H_2O \xrightarrow{\Delta} Ca(OH)_2 + CO_2 + 2NH_3$

SECTION – III
Reasoning Type

9. **(b)** 

Driving force of reaction is provided by the measure of relief in bond strain on going from carbonyl compound (where C–C–C bond angle is 60° compared to normal sp^2 value of 120°) to hydrate (C–C–C bond angle is 60° compared to normal sp^3 value of 109°).

10. **(a)** $Hg_2(NO_3)_2 \longrightarrow Hg_2^{2+} + 2NO_3^-$

11. **(b)** At 25°C, the ionic product of H_2O and D_2O are 10^{-14} and 3×10^{-15} respectively justifying that dissociation of H_2O is much more than that of D_2O giving more H^+ ions than D^+ ons.

12. **(b)** Due to low K_{sp} of $BaSO_4$ in water, it is nearly insoluble in water. So, it is not absorbed by body and we do not get any toxicity.

SECTION–IV
Comprehension Type

Passage-I

13. **(a)** Molecular weight of (A) = $\frac{0.369 \times 22400}{67.2} = 123$

Let bromo derivative be R–Br.

Molecular weight of R = 123 – 80 = 43

$C_nH_{2n+1} = 43$; $n = 3$

$\therefore$ Molecular formula = **C_3H_7Br**.

14. **(d)** (A) is C_3H_7Br which may exist as $CH_3CH_2CH_2Br$ or $CH_3CH(Br)CH_3$.

$CH_3CH_2CH_2Br$ (A) $\xrightarrow{NaOH}$ $CH_3CH_2CH_2OH$ (B)

- $\xrightarrow[250°C]{Alumina}$ $CH_3CH_2CH_2OCH_2CH_2CH_3$ (C)
- $\xrightarrow[350°C]{Alumina}$ $CH_3CH=CH_2$ (D)

$$\underset{\text{(D)}}{CH_3CH=CH_2} \xrightarrow{HBr} \underset{\text{(Isomer of A)}}{CH_3CH(Br)CH_3} \xrightarrow{\text{Alco. KCN}} CH_3-\underset{\underset{CN}{|}}{CH}-CH_3 \xrightarrow[-NH_3]{2H_2O/H^+} \underset{\underset{\text{(E)}}{COOH}}{CH_3-\underset{\underset{|}{}}{CH}-CH_3}$$

$$\underset{\underset{\text{(E)}}{COOH}}{CH_3-\underset{\underset{|}{}}{CH}-CH_3} \xrightarrow{Br_2/\text{Red P}} \underset{\underset{\text{(Optically inactive)}}{COOH}}{CH_3-\underset{\underset{|}{}}{C(Br)}-CH_3}$$

If (A) would be $CH_3–CH(Br)–CH_3$ then isomer of (A) is $CH_3–CH_2–CH_2–Br$ and (E) is $CH_3–CH_2–CH_2–COOH$ which would give optical active compound $CH_3–CH_2–CH(Br)–COOH$ on its HVZ reaction.

15. **(d)** $CH_3CH_2CH_2OCH_2CH_2CH_3 \xrightarrow{HBr} \underset{\text{(A)}}{CH_3CH_2CH_2Br} + \underset{\text{(B)}}{CH_3CH_3CH_2OH}$

Passage-II

16. **(c)** $P = \dfrac{829 \text{ mm of Hg}}{760 \text{ mm of Hg}} = 1.09 \text{ atm}$

Mole of N_2 required $= \dfrac{PV}{RT} = \dfrac{(1.09 \text{ atm})\,(45.5\,L)}{(0.0821\,L \text{ atm } K^{-1} \text{ mol}^{-1})\,(295\,K)} = 2.05 \text{ mol of } N_2$

$2NaN_3(s) \longrightarrow 2Na + 3N_2(g)$

3 mole of N_2 are produced from = 2×65 g of NaN_3

2.05 mole of N_2 are produced from = $\dfrac{2 \times 65}{3} \times 2.05$ = **88.8 g NaN_3**.

17. **(a)** Applying $\dfrac{P_1V_1}{T_1} = \dfrac{P_2V_2}{T_2}$

$V_2 = V_1 \times \dfrac{P_1}{P_2} \times \dfrac{T_2}{T_1}$

$= 4.19 \times 10^3 \times \dfrac{754}{76} \times \dfrac{240}{295.5}$ = **3.38×10^4 L**.

18. **(d)**

SECTION–IV
Matrix–Match Type

1. (A) – 1, 3, 4 ; (B) – 2, 3 ; (C) – 4, 5 ; (D) – 2, 3, 4

Sol of $Fe(OH)_3$ containing $FeCl_3$ will form positive colloid [i.e. $Fe(OH)_3.Fe^{3+}$] which can be formed by peptization and chemical method.

By hydrolysis: $FeCl_3 + 3H_2O \longrightarrow \underset{\text{(Collidal sol)}}{Fe(OH)_3} + 3HCl$

$Fe(OH)_3$ sol can be obtained by boiling solution of its chloride ($FeCl_3$).

A precipitate of silver iodide can be peptized by shaking with a dilute solution of KI and due to excess of I^- ion it will be negative colloid which may be represented as $AgI.I^-$.

Colloidal sol of gold metal can be obtained by Bredig Arc method and by reduction of very dilute solution of their salts with a suitable reducing agents e.g.

$$2AuCl_3 + 3SnCl_2 \longrightarrow \underset{\text{(Gold sol)}}{2Au} + 3SnCl_4$$

As_2S_3 in H_2S in negative collid due to excess of sulphuide ion, chemically obtained by double decomposition.

$$As_2O_3 + 3H_2S \longrightarrow \underset{\text{(Collidal sol)}}{As_2S_3} + 3H_2O$$

2. (A) – 1, 2 ; (B) – 1, 2, 3 ; (C) – 2, 4, 5 ; (D) – 2

State functions are those vairables whose value depends on initial and final state of system.

Extensive property are those whose value depends on amount of substance.

$dq = du + dw$

$TdS = d(H - PV) + PdV$; $TdS = dH - PdV - VdP + PdV$; $TdS + VdP = dH$

At constant P, $TdS = dH$

$$T = \left(\frac{dH}{dS}\right)_P$$